

Nathan Brewer

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Portfolio: natebrewer.github.io

EDUCATION

Swarthmore College

Swarthmore, PA

Bachelor of Science in Engineering, Minor in Applied Mathematics

Expected May 2028

- Major GPA: 3.96, Cumulative GPA: 3.89
- Relevant Coursework: Mechanics of Solids, Thermofluid Mechanics, Linear Physical Systems Analysis, Principles of Computer Engineering, Electrical Circuit Analysis
- Honors and Memberships: Sigma Xi, American Society of Mechanical Engineers

WORK EXPERIENCE

Research and Development Intern

May 2026 – August 2026

Ryzing Technologies

Staunton, VA

- Led three design iterations of tent circumferential straps for US Army Arctic Medical Operations, increasing tensile strength by 15% and standardizing textile dimensions for manufacturing; personally sewed production units
- Validated roll MOI of USMC vehicle model to within 1% of DOD datasheet specifications using MATLAB analysis and lead-weight ballasting, enabling wave pool roll testing
- Automated in-house manual wind testing setup with a servo-driven, PID-controlled mechatronic system running programmable RPM profiles from CSV at 2% tolerance, enabling repeatable, client-customizable wind testing
- Developed supplier evaluation matrices and coordinated directly with vendor representatives to source EMI shielding fabric for a secure communications program, achieving 70 dB average attenuation
- Produced dimensioned drawing packages with tolerances and datum references across all design projects

Engineering Research Associate

October 2024 – Present

Swarthmore College

Swarthmore, PA

- Built a biologically accurate robotic barnacle nauplius model, enabling University of Washington collaborators to study the effects of morphological variation on hydrodynamic performance
- Designed a fully modular drive assembly with three appendage levels, allowing investigation of vertical positioning of appendage sets
- Developed a light-activated motor control circuit with adjustable threshold and drive voltage, providing locomotion across a range of speeds and ambient lighting conditions
- Consolidated actuation from three independent drive motors to a single common drive shaft, reducing mechanical complexity and power draw while maintaining 5% angular sweep accuracy
- Reduced mechanism height by 20% relative to prior designs, improving profile accuracy for fluid testing

Physics Teaching Assistant

September 2025 – December 2025

Swarthmore College

Swarthmore, PA

- Selected to lead weekly help sessions on mechanics topics (kinematics, dynamics, energy/momentum) for 60+ students in General Physics I

EXTRACURRICULAR ACTIVITIES

Swarthmore Robotics

- Designed and fabricated weapon, armor, and control/drive systems of 1 lb. BattleBot
- Use Fusion360 and ANSYS to run dynamic impact simulations for weapon and armor designs
- Modeled custom second-order hyperelastic TPU material for armor design in ANSYS
- Leading development and fabrication of NHRL-compliant Beetleweight (3 lb.) combat robot

Model Rocketry

- Designed a custom layup and curing setup to hand-wrap a carbon fiber body tube
- Model rocket designs in SolidWorks and OpenRocket to optimize stability margin
- Machined tooling to create bentonite clay nozzle geometry, achieving ~115 second specific impulse
- Achieved ~20% burn-rate increase in home-cast KnSu rocket motors through iterative testing of iron oxide doping mixtures

SKILLS

Design: SolidWorks, Fusion360, Onshape, ANSYS Mechanical, SimuLink, KiCad, MultiSim

Computer Languages: MATLAB, Python (SciPy, NumPy, Matplotlib), SQL

Fabrication: MIG/TIG welding, manual lathe, manual mill, basic carpentry, 3D printing, soldering, oscilloscopes